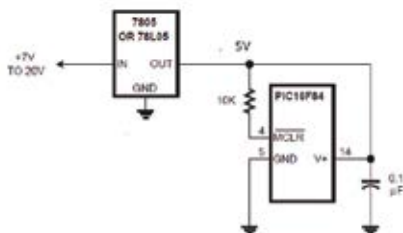


mounted close to the PIC, to protect the PIC and other components from electrical noise. This capacitor should always be used, because you can't be sure that the power supply available to you doesn't have any AC component present in it.

The MCLR pin, which stands for Master CLeaR, is a pin which is used to 'reset' the chip, which means clearing the data registers of PIC. This pin is active low, which means that it gets active and performs its function when provided with low voltage (0V), and is passive when supplied with high voltage (5V). Normally it is connected to 5V through a 10k resistor.

Grounding MCLR will clear RAM and reset the PIC. It is always a good practice to ground the MCLR momentarily as you switch on your circuit, so as to be sure that the circuit doesn't have any 'memory' of its previous operation.



Clock requirements

All CPUs need a clock so that they can time their instructions. For example, if you want a particular instruction to be executed 5 seconds after the last instruction is executed, you have to provide some timing device which the processor will use to know when those 5 seconds get over. This timing device is called 'clock' and its frequency is called 'clock frequency'.

The higher the clock frequency, higher is the speed of processor. Generally, all processors have a maximum clock frequency which can be provided to them, in case of 16F84 it is 4MHz. Also, there is no lower limit to the clock frequency. The lower clock you provide, slower will be the processor, and lesser power it will consume.

There are two commonly used methods to provide clock:

1. Using a resistor and a capacitor: It makes use of inexpensive capacitor and resistor, but the clock may sometimes not be accurate, so you should avoid using this method when you need very accurate clock.

2. Using Oscillator crystal: A crystal is another electronic component, which is cheap and provides very accurate clock. We would be using this method for our robot. 